

STEP 9

CI/CD with GitHub Actions

Every push is automatically trained, tested, drift-checked, containerised and smoke-tested.

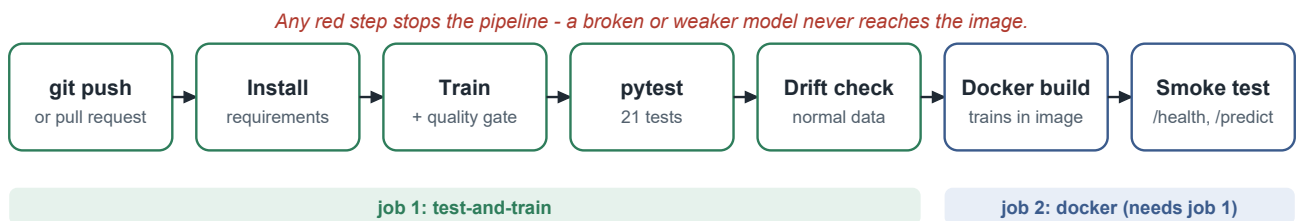


Figure 5 - The workflow in `.github/workflows/ci.yml`. Job 2 only starts if job 1 passes.

Steps

Step	Command	Fails when
Install	<code>pip install -r requirements.txt pytest httpx</code>	A dependency cannot be installed
Train	<code>python -m foodwaste.train</code>	Data is invalid or macro F1 < 0.80
Test	<code>pytest -v</code>	Any of the 21 tests fails
Drift check	<code>python -m foodwaste.drift --simulate normal --fail-on-drift</code>	Normal data is flagged (a false alarm means the check is broken)
Upload	<code>actions/upload-artifact</code>	- (saves metrics, confusion matrix, reports)
Docker build	<code>docker build -t food-waste-api .</code>	Image cannot be built or training inside fails
Smoke test	<code>docker run + curl /health + curl /predict</code>	Container does not start or answer

Put it on GitHub

```

cd C:\Users\priya\MLOps\food-waste-mlops
git init
git add .
git commit -m "Food waste MLOps project"
git branch -M main
git remote add origin https://github.com/<your-username>/food-waste-mlops.git
git push -u origin main
  
```

Open the repository on GitHub and go to the **Actions** tab to watch the run. A green tick means every gate passed.

Why it matters

This is the 'Ops' in MLOps: no manual step decides whether a model is good enough. The same automatic checks run for every change, by anyone, on a clean machine.